Electric Circuits

Write the numbers 1 – 10 into your book.

THE NEW UNIT

Find out which harder words in the new unit you can spell.

Review: Swap books with the person next to you and mark their answer with a red pen. Correct any wrong spelling.

1. Electrons
2. Diode
3. Resistance
4. Potential
5. Component
6. Thermistor
7. Electricity
8. Ammeter
9. Voltmeter
10. Coulombs

Electric circuits

Learning Objective:

1. Know circuit symbols.

2. Recall that current is a rate of flow of electric charge.

3. Recall that current (I) depends on resistance (R) and potential difference (V).

4. Explain how an electric current passes round a circuit.

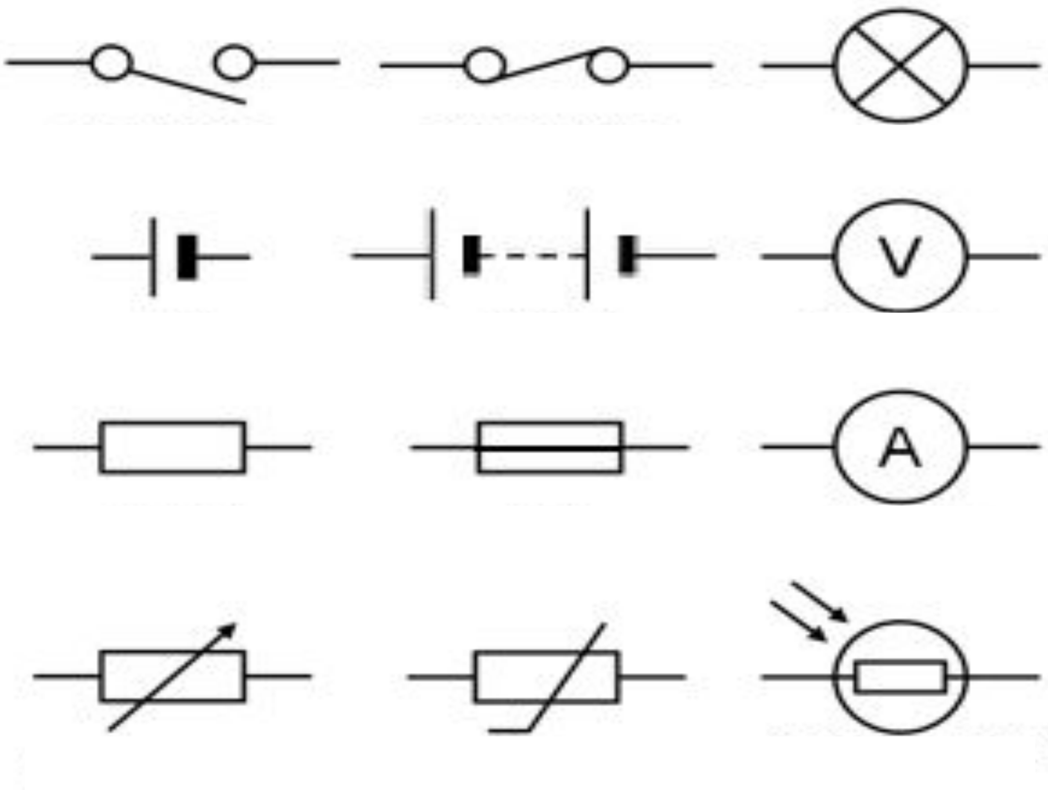
Queen Elizabeth School

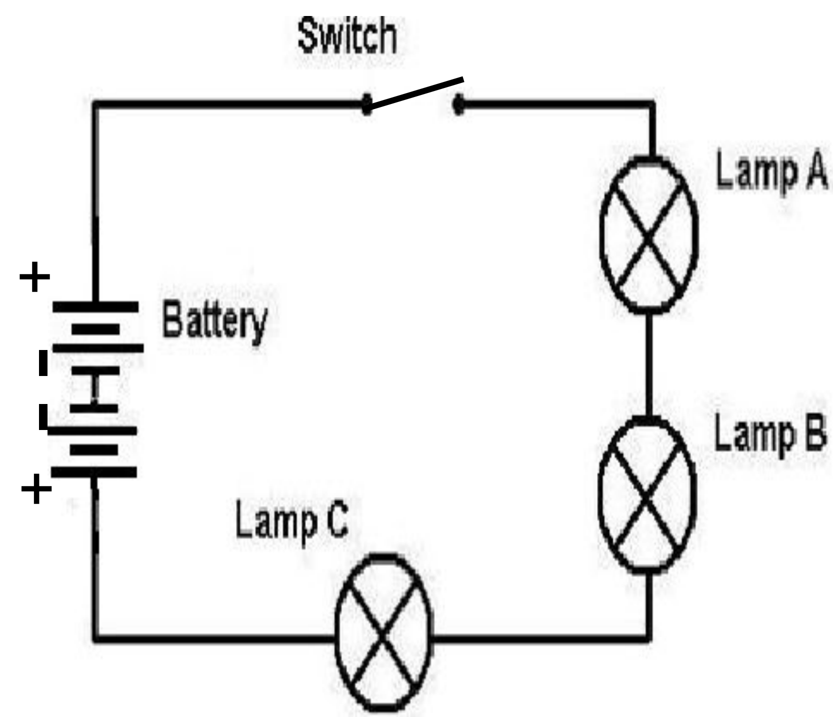
Current and Charge *

Key Words: Current, charge, coulombs, component, electrons

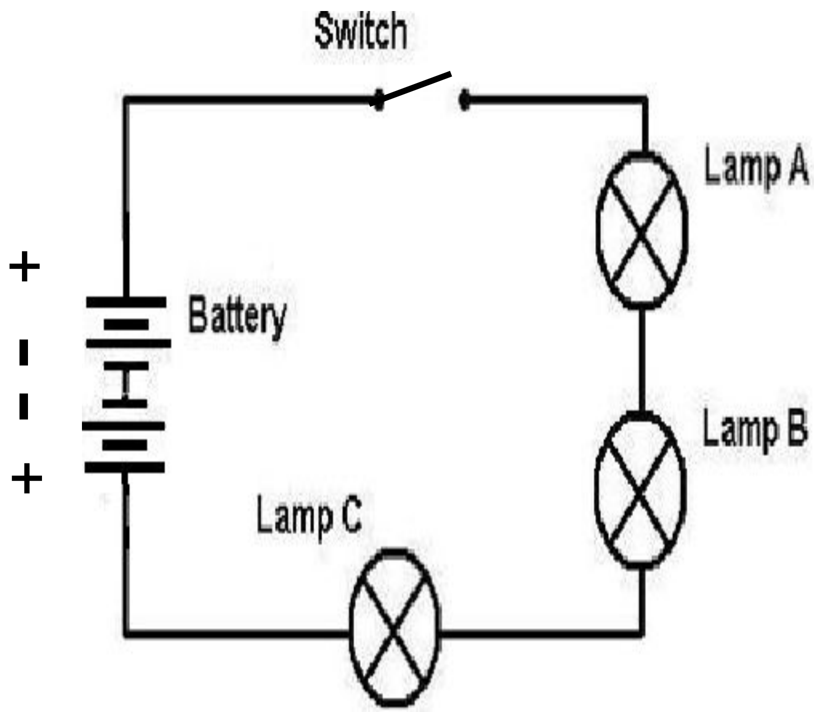
<u>DO NOW</u>: Do you know any of these electric circuit symbols? Draw and name the ones you know.

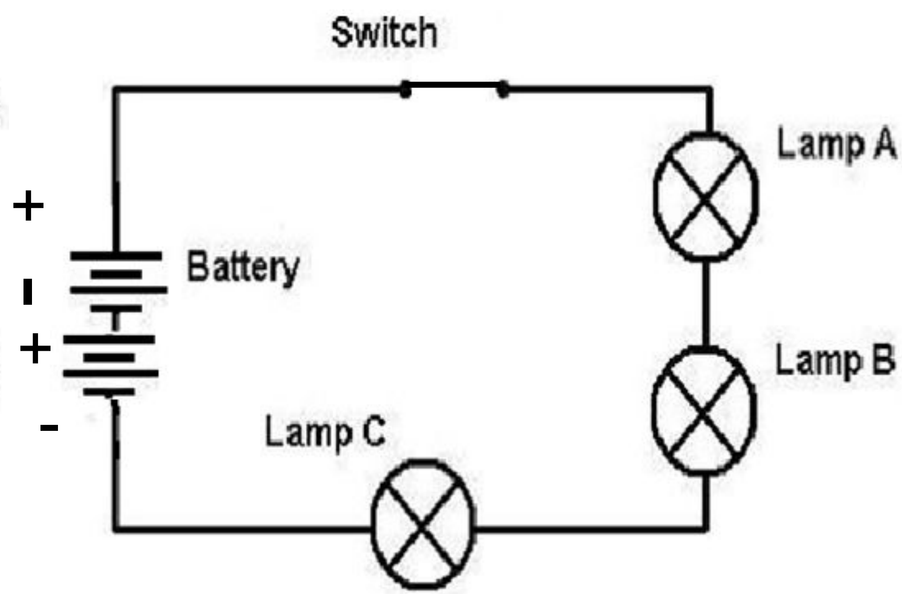
<u>EXT</u>: State two reasons this circuit won't work. Draw the corrected





Electric current can
Only flow from positive
To negative



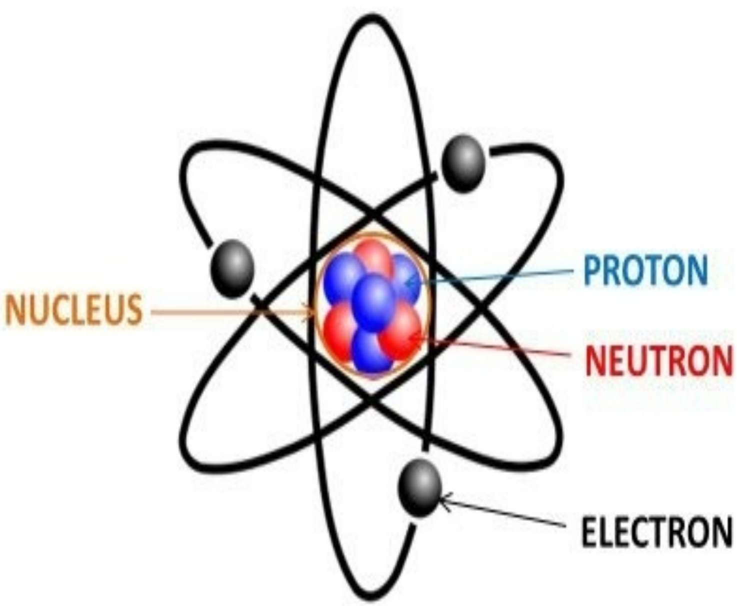


LOb: Understand the terms current and charge.

Keywords: Current, charge, coulombs, component, electrons

<u>Success Criteria:</u>

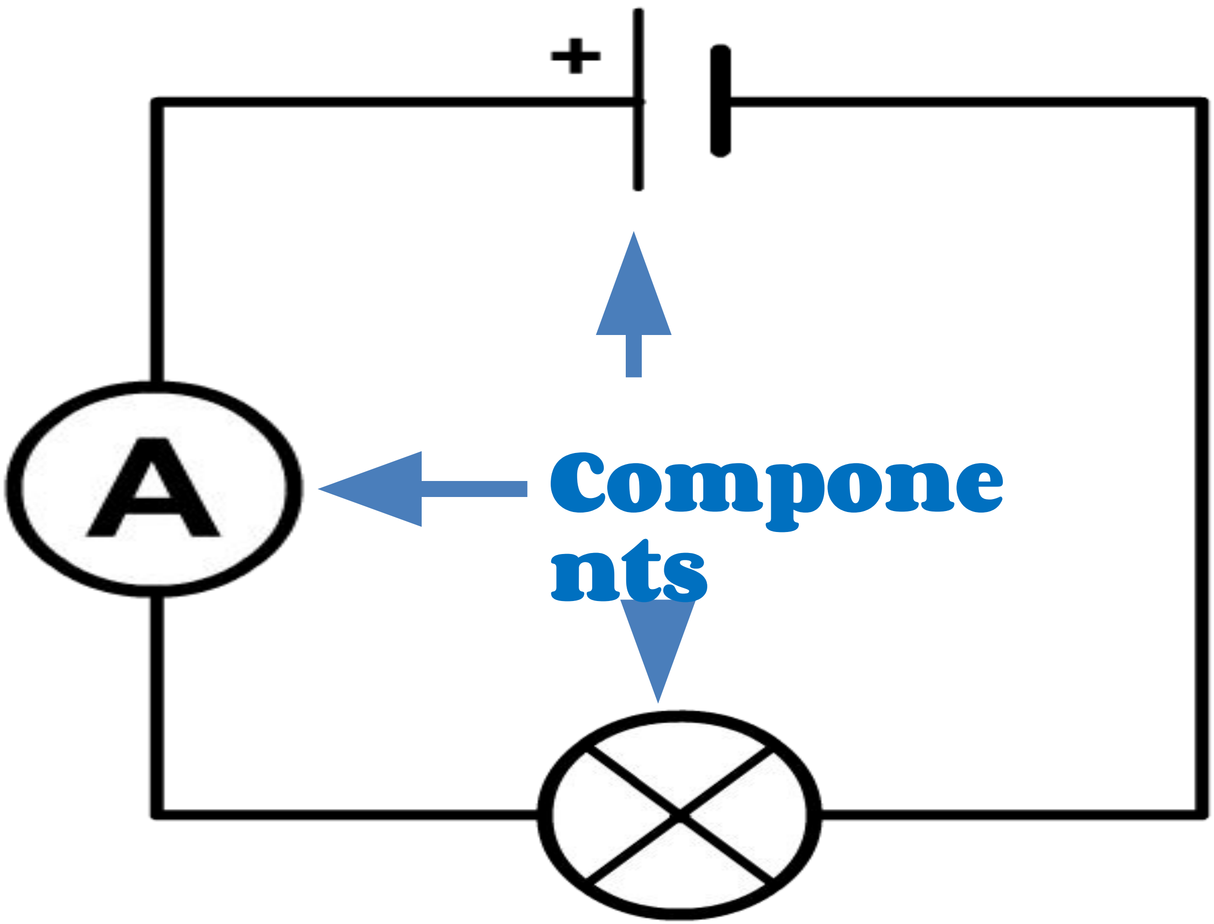
- Recall how electric circuits are shown as diagrams
- Describe the current in an electric circuit
- Calculate the size of an electric current



Did you know… electricity is all to do with how electrons move through wires.

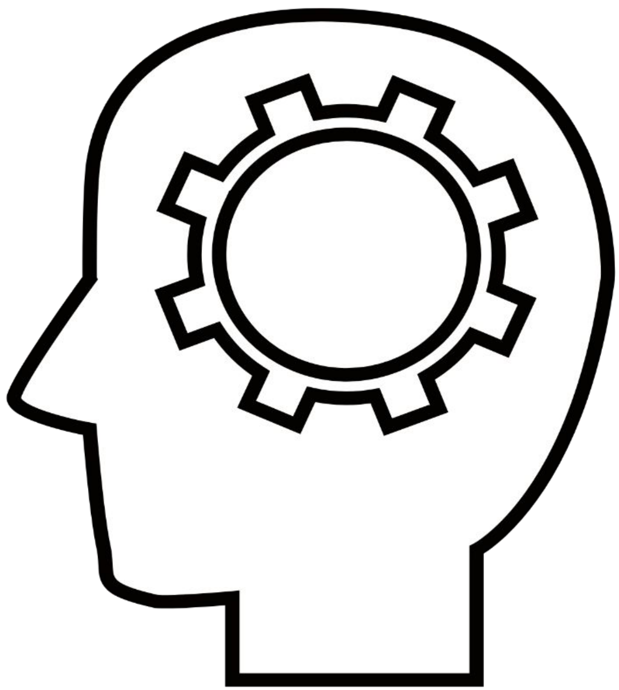
This is a **circuit diagram**. It shows you in the simplest way how the **components** are connected in a circuit.

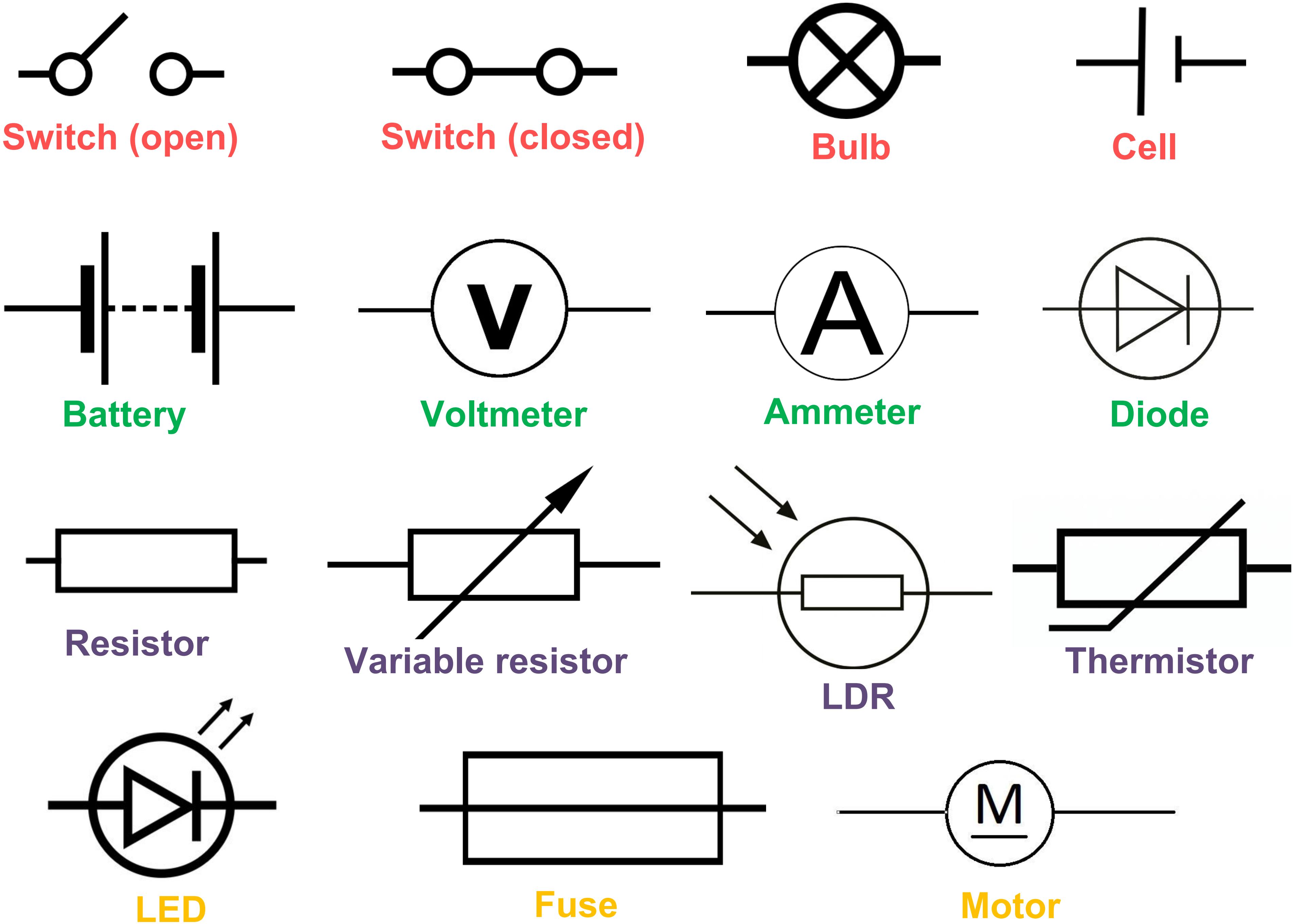
Each component has its own symbol. at GCSE you need to recognise them and remember what they are used for.



Activity: On the next slide you will see all the components you need to know. You have 45 seconds to recognise them.

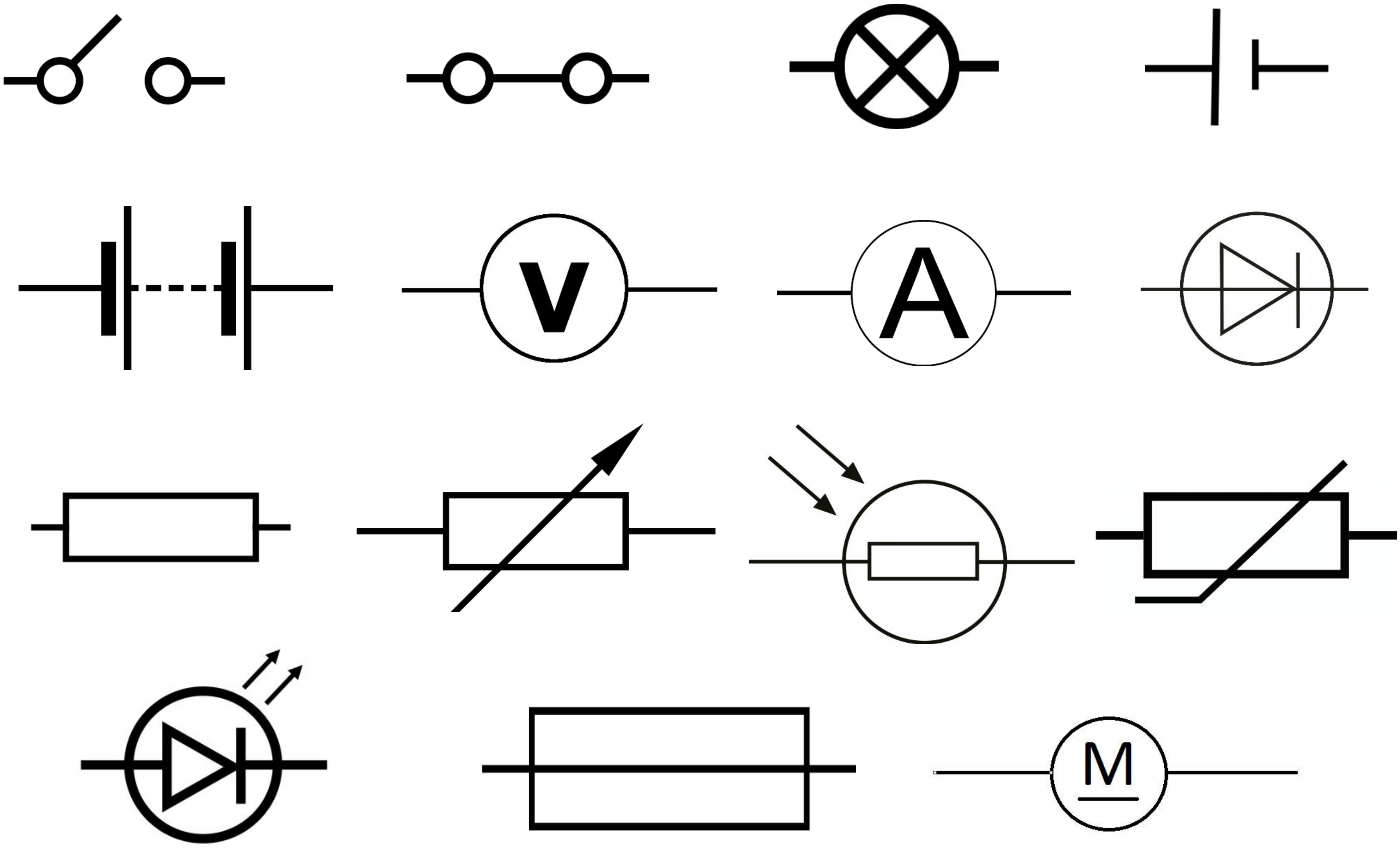
MEMORY TEST




Switch (open)
Switch (closed)
Bulb
Cell
Battery
Voltmeter
V
Ammeter
A
Diode
Resistor
Variable resistor
LDR
Thermistor
LED
Fuse
Motor
M

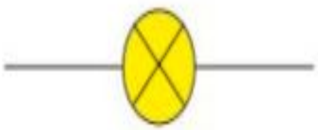
Review: Can you remember the symbols?

Complete the names onto your sheet.

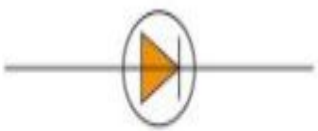


A cell is necessary to push electrons around a complete circuit. A battery consists of two or more cells. The + symbol next to the long line of the cell indicates that this is the positive terminal of the cell.

A switch enables the current in a circuit to be switched on or off.



An indicator, such as a bulb, is designed to emit light as a signal when a current passes through it.

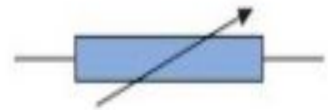


A diode allows current through in one direction only.

A light-emitting diode (LED) emits light when a current passes through it.

An ammeter is used to measure electric current.

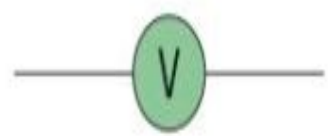
A fixed resistor limits the current in a circuit.



A variable resistor allows the current to be varied.

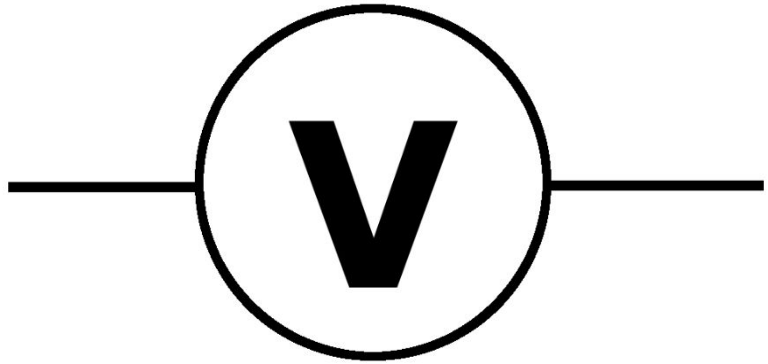
A fuse is designed to melt and therefore 'break' the circuit if the current through it is greater than a certain amount.

A heater is designed to transfer the energy from an electric current to heat the surroundings.

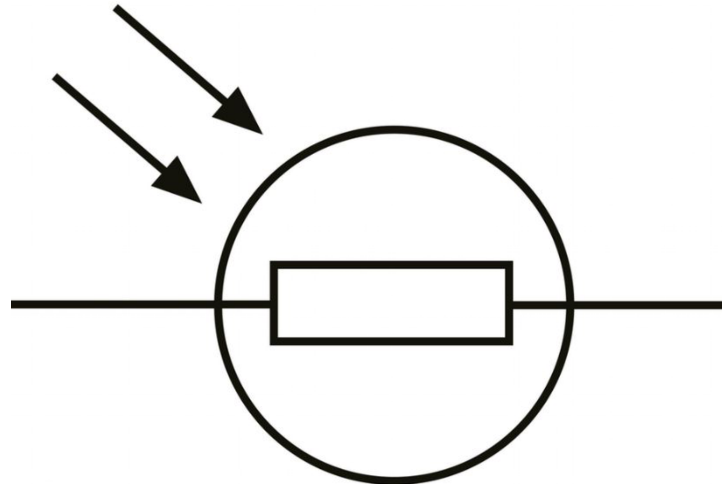


A voltmeter is used to measure potential difference (i.e. voltage).

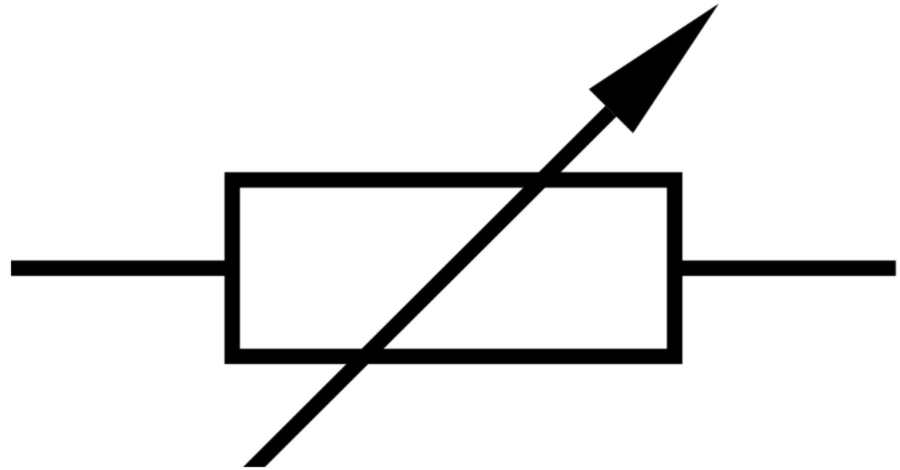
Bul
b



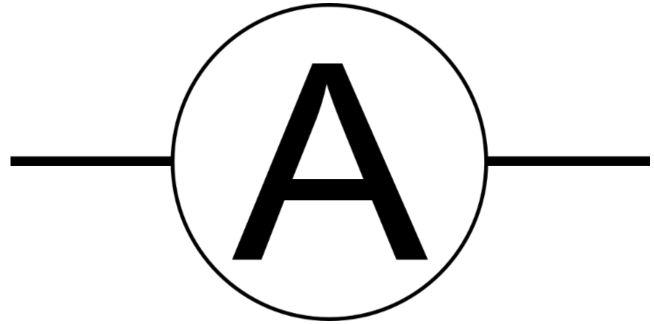
Voltmet
er



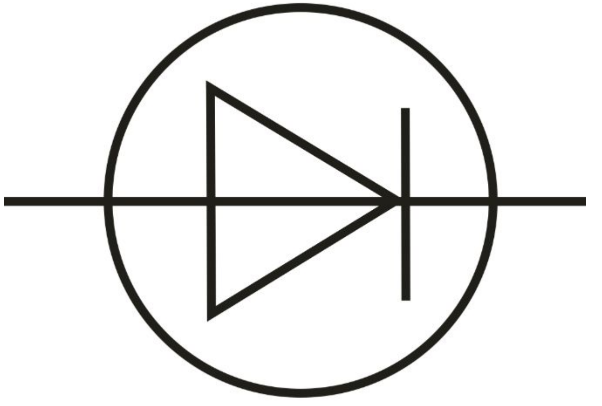
LD
R



Variab
le
Resist

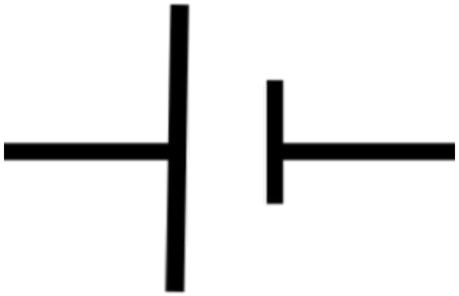


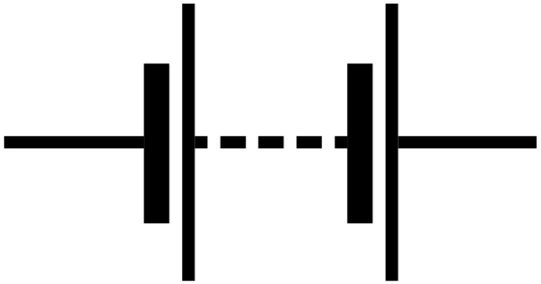
Ammeter



Dio
de

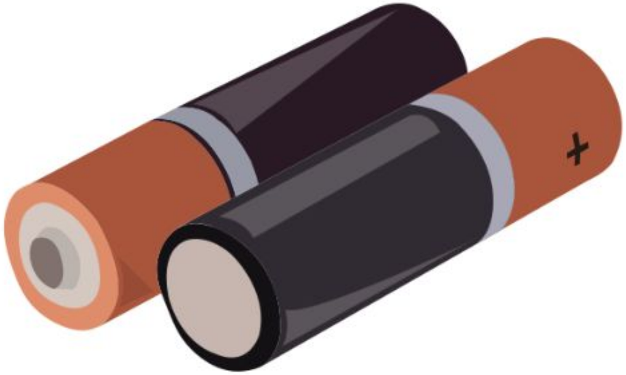
A cell is required to push electrons round a circuit. A battery is two or more cells.





Cell

Battery

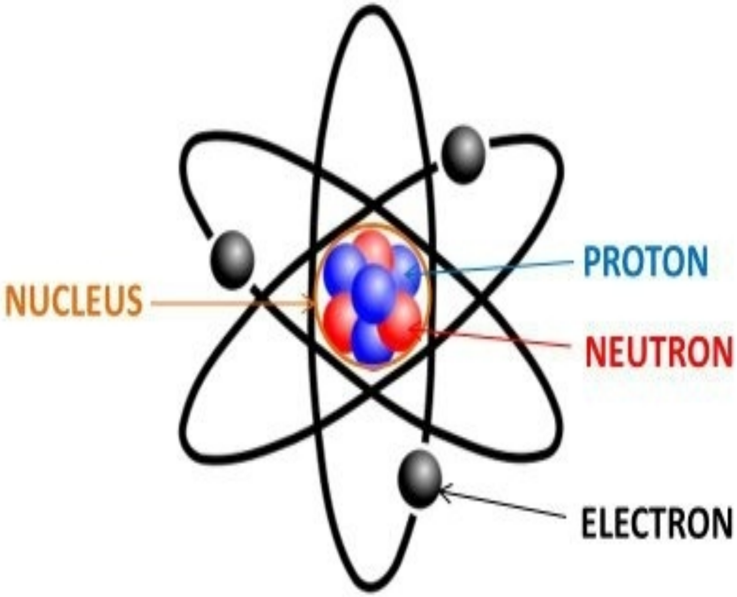


LOb: Understand the terms current and charge.

Keywords: Current, charge, coulombs, component, electrons

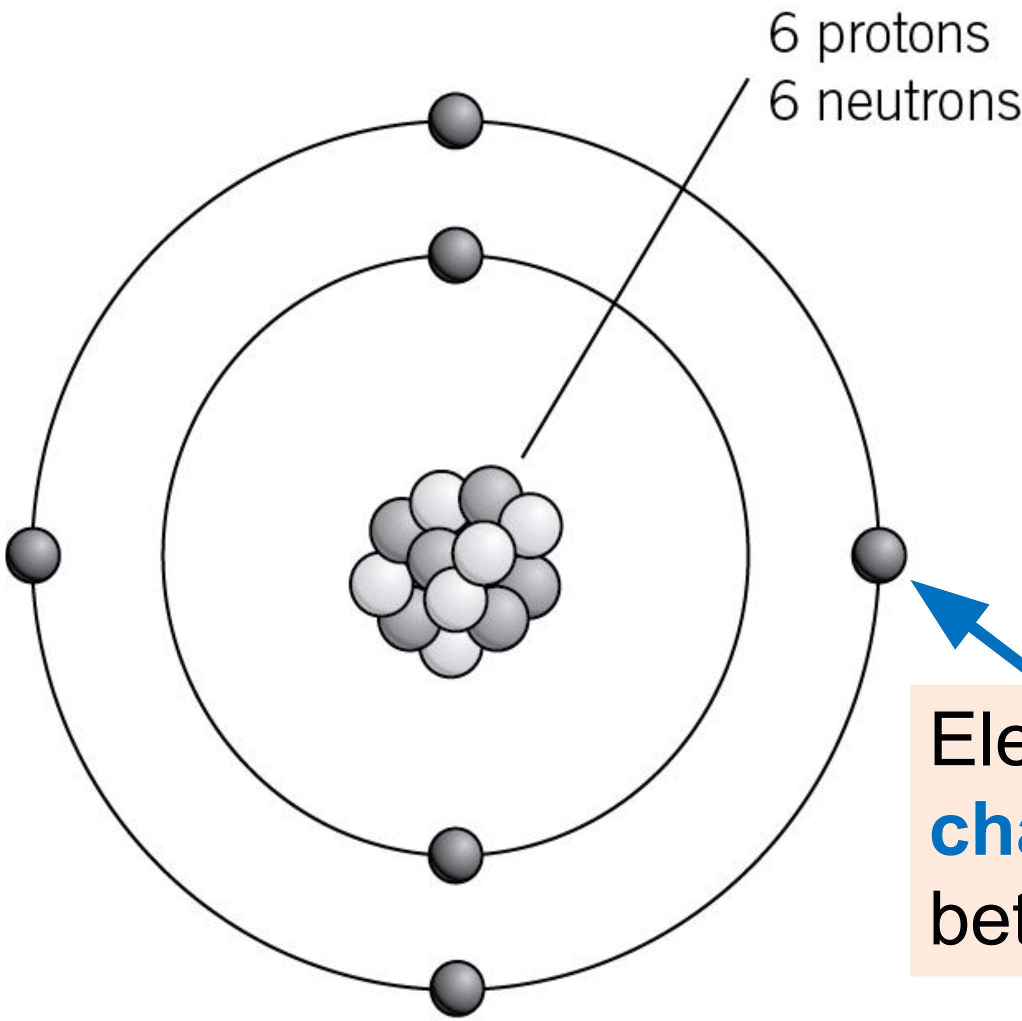
Success Criteria:

- Recall how electric circuits are shown as diagrams
- Describe the current in an electric circuit
- Calculate the size of an electric current



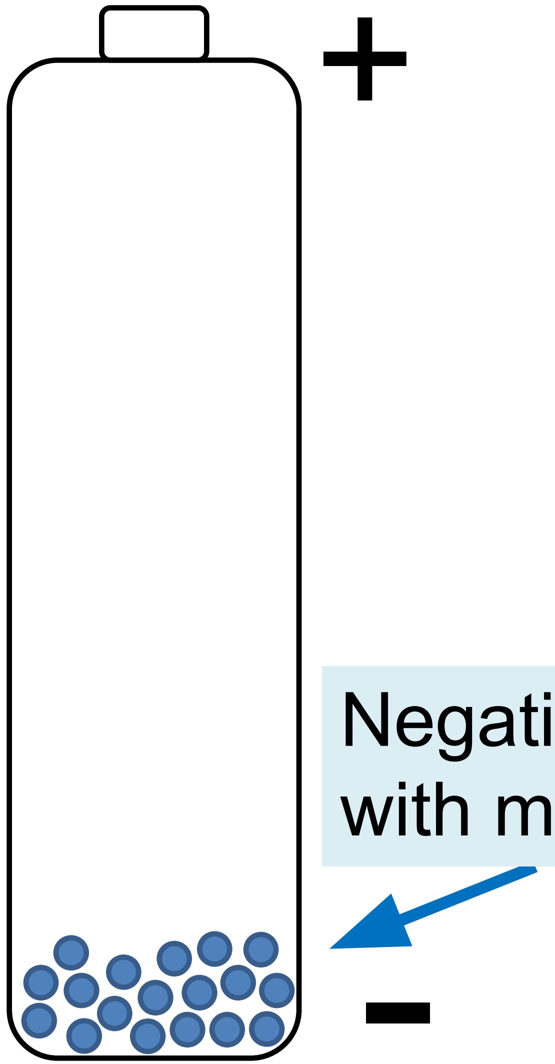
Did you know… electricity is all to do with how electrons move through wires.

To understand electricity we need to understand **charge**.



Electrons have a **negative charge.** They like to move between atoms.

In a cell a chemical reaction produces more electrons at one end.



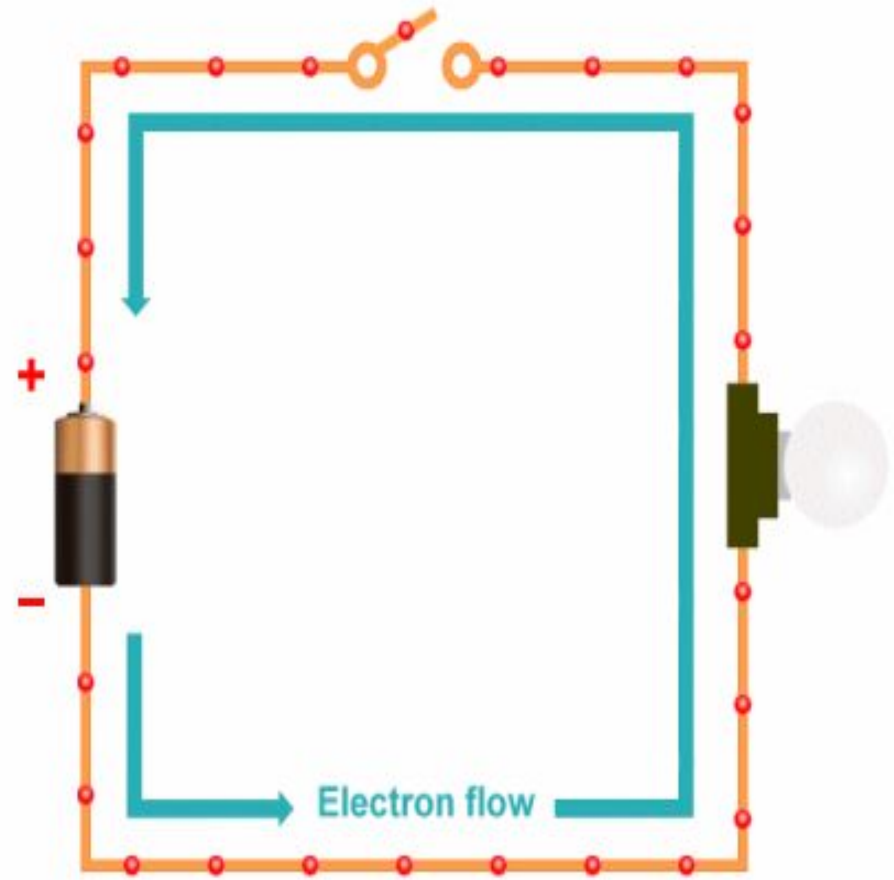
Negative end of the battery with millions of electrons.

What is electric current?

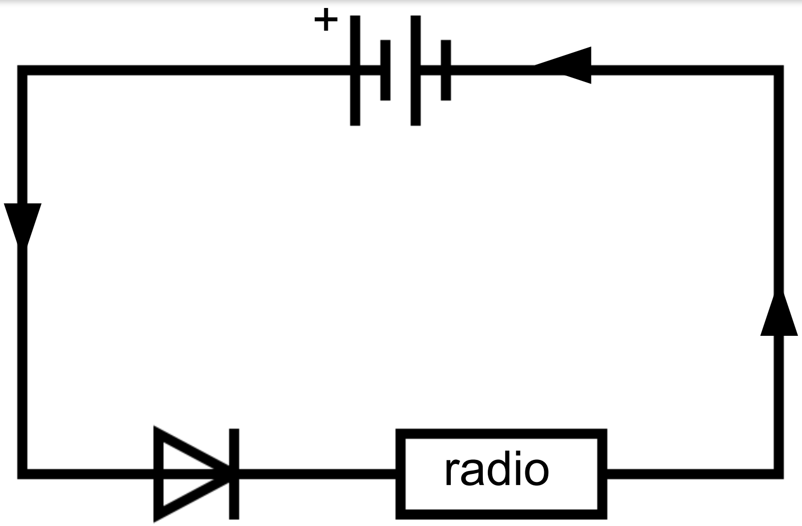
When the battery is connected in a complete circuit the **electrons flow** from the negative to the positive. This is a flow of charge (negative electrons) which is called...

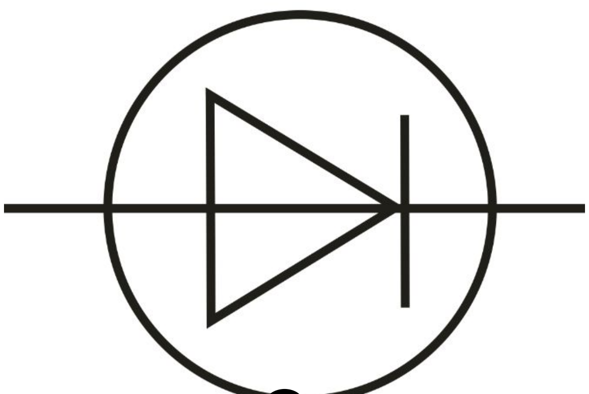
Current	A flow of electric charge.

Note. *Though the **electrons flow** from negative to positive we show **current flow** as going from positive to negative.*



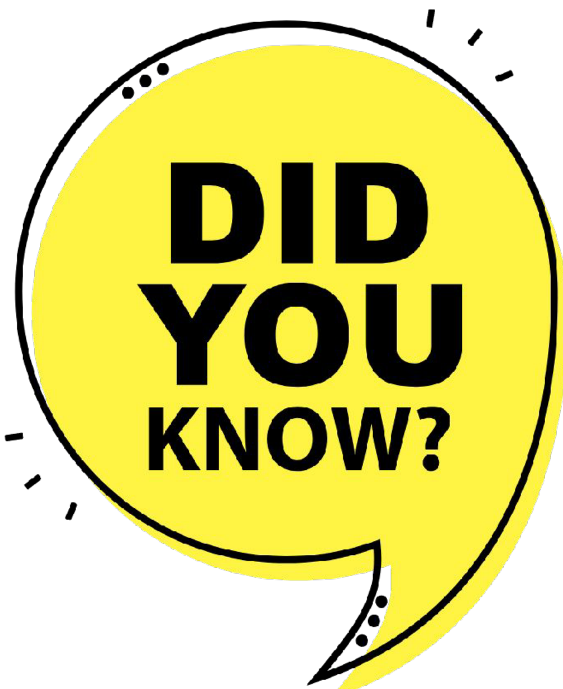
If you put batteries in the wrong way in a portable radio it could damage it if it wasn’t for the diode.



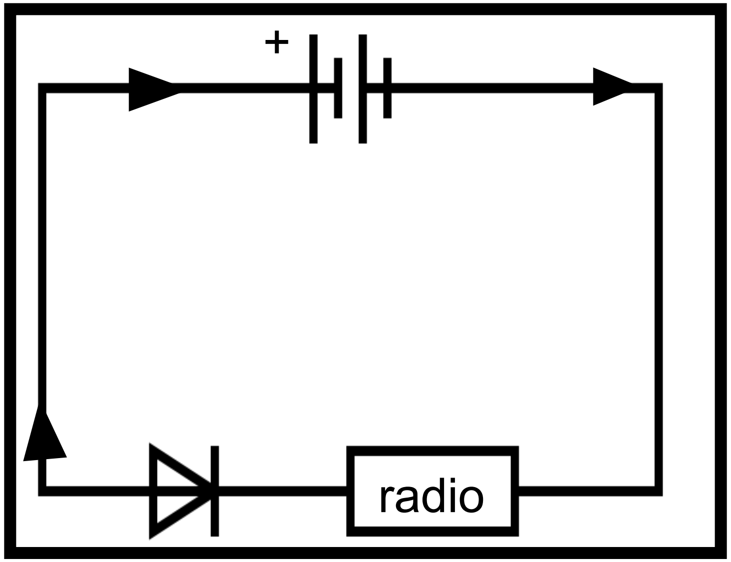


Dio

Stops current going the wrong way.

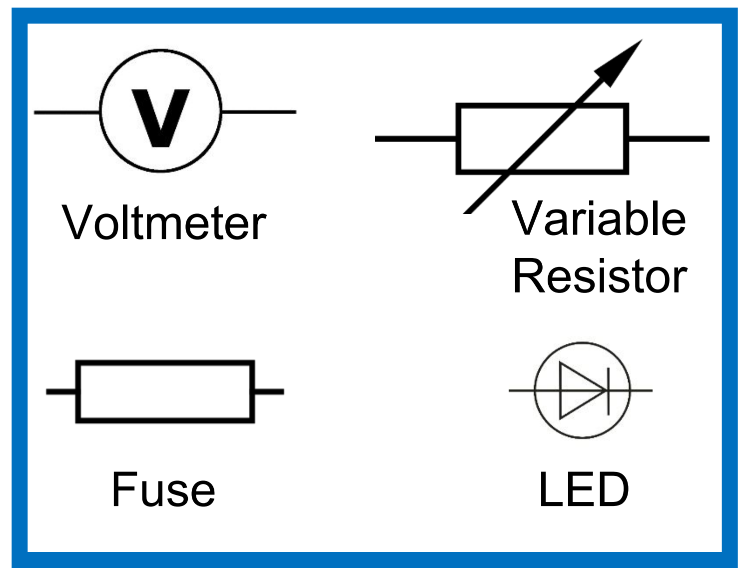


Activity: State what is wrong with each.

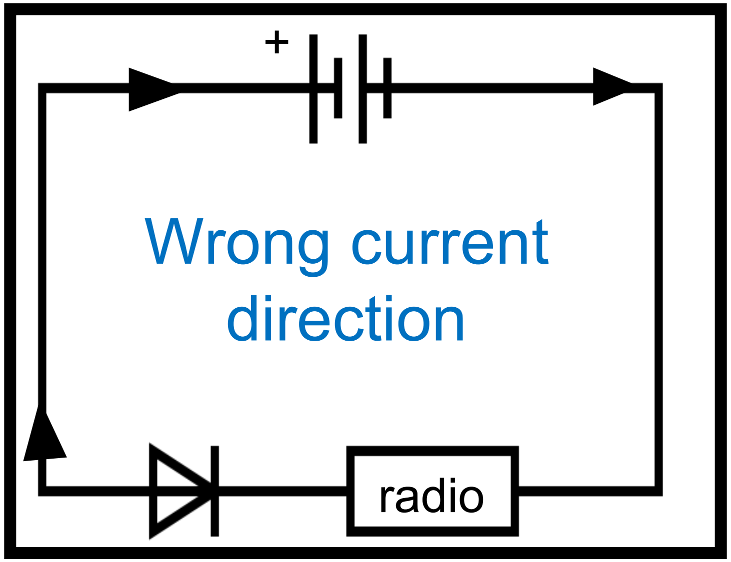


Current | A flow of electron charge.

e^+



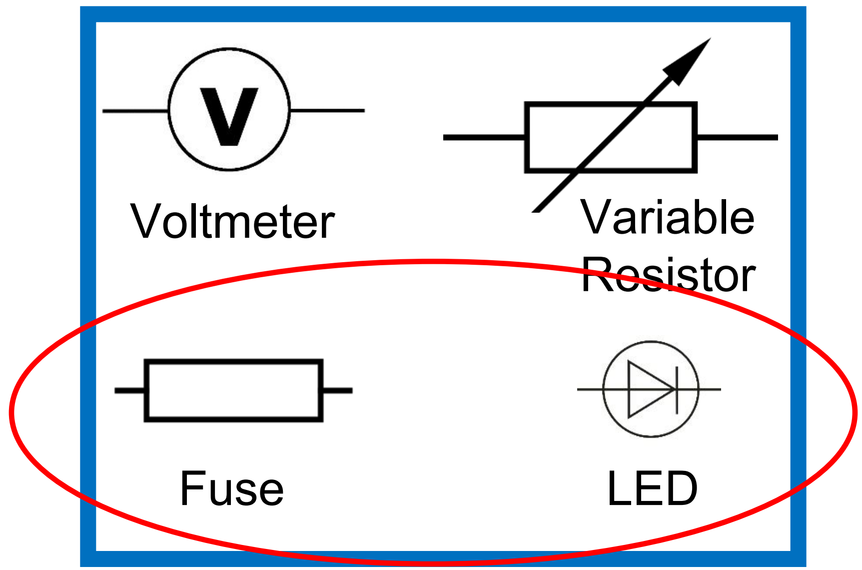
Review



Current | A flow of ~~electron~~ charge.

Should be e^-

e^+



LOb: Understand the terms current and charge.

Keywords: Current, charge, coulombs, component, electrons

Success Criteria:

- Recall how electric circuits are shown as diagrams
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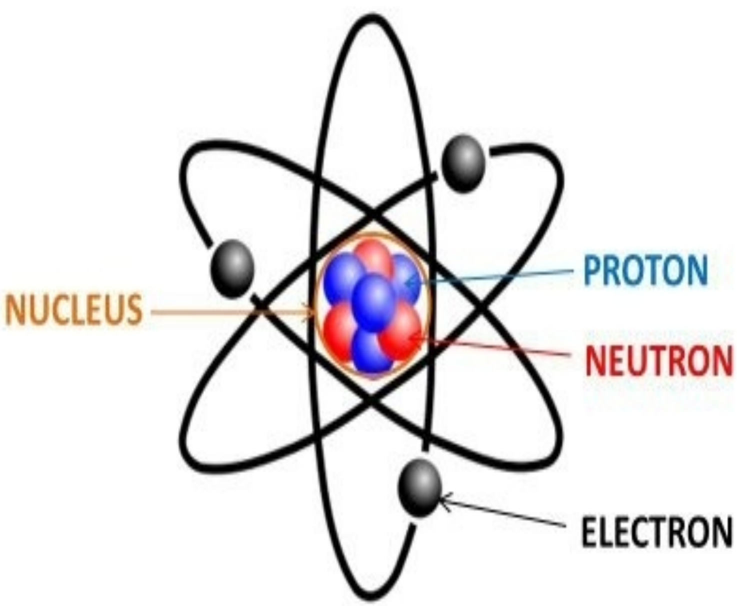


Did you know… electricity is all to do with how electrons move through wires.

Activity: Watch this video make three bullet points of what you have learnt.

- ❑
- ❑
- ❑

WATCH AND

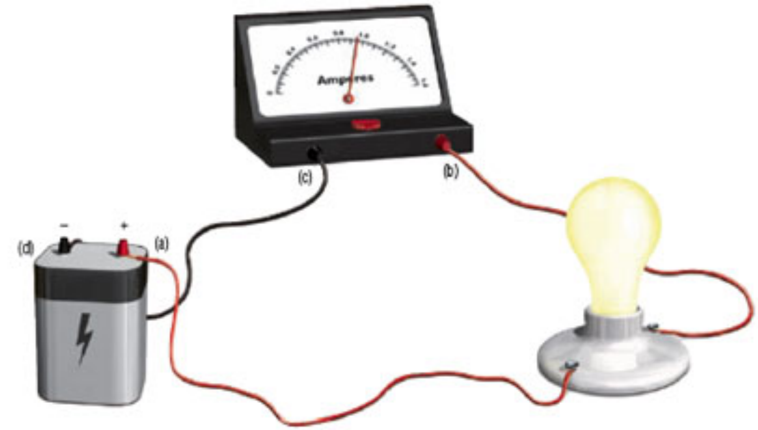
LEARN

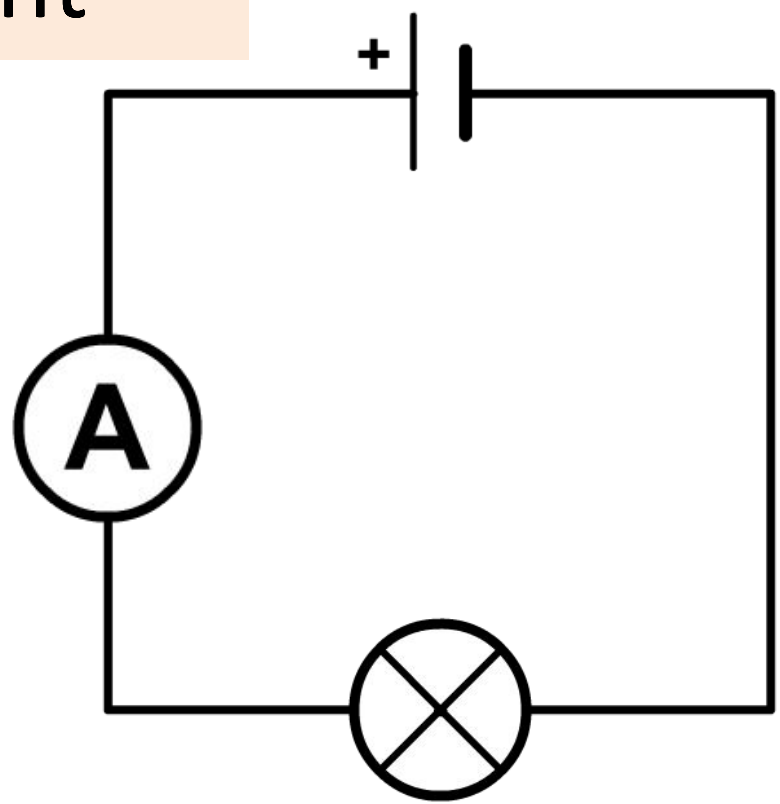
AFTERALL KNOWLEDGE IS

POWER!

Current is measured in amps (A) using an ammeter.

The ammeter is attached in series with every other component

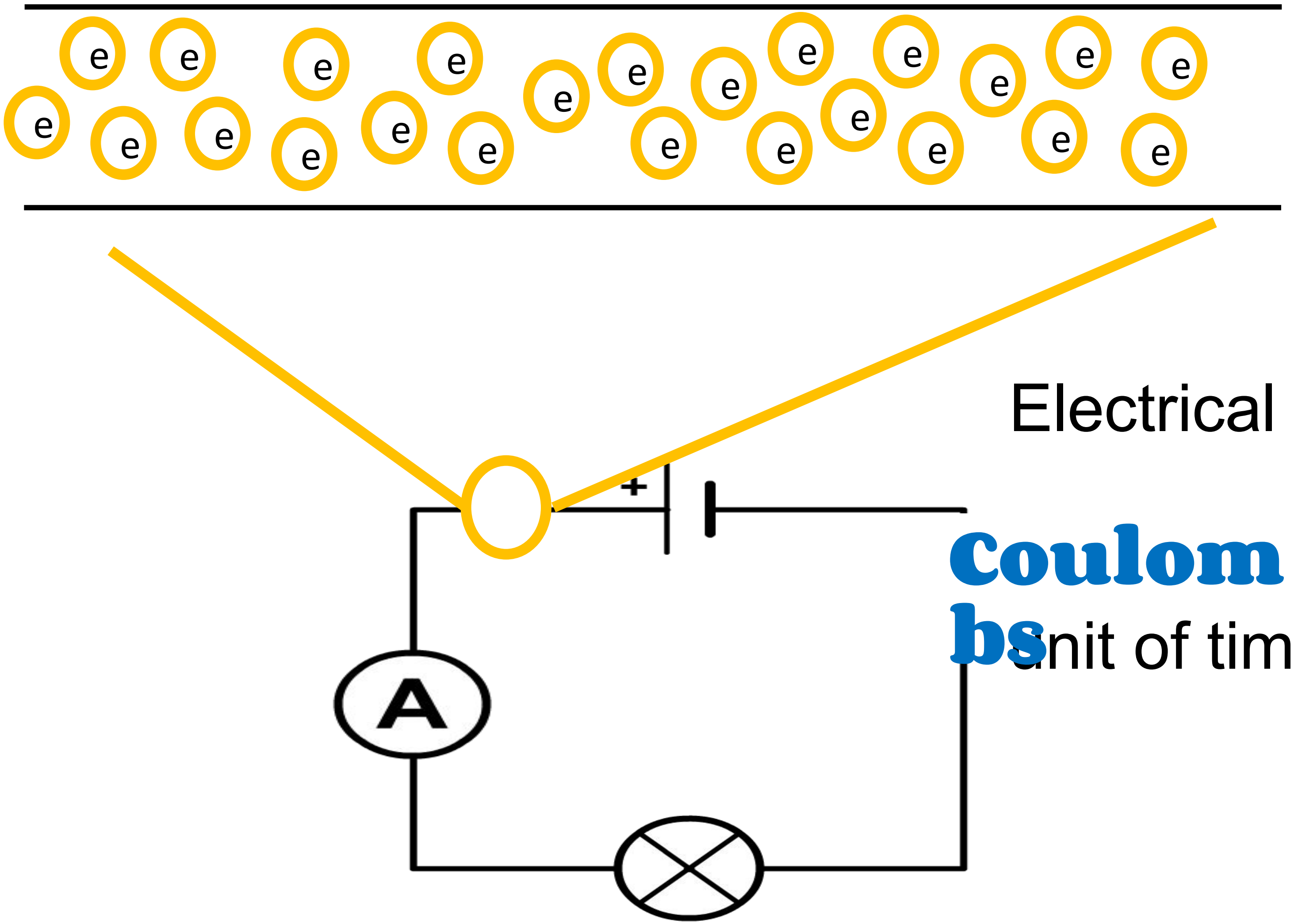




Electrical current is:

“electrons per unit of time”

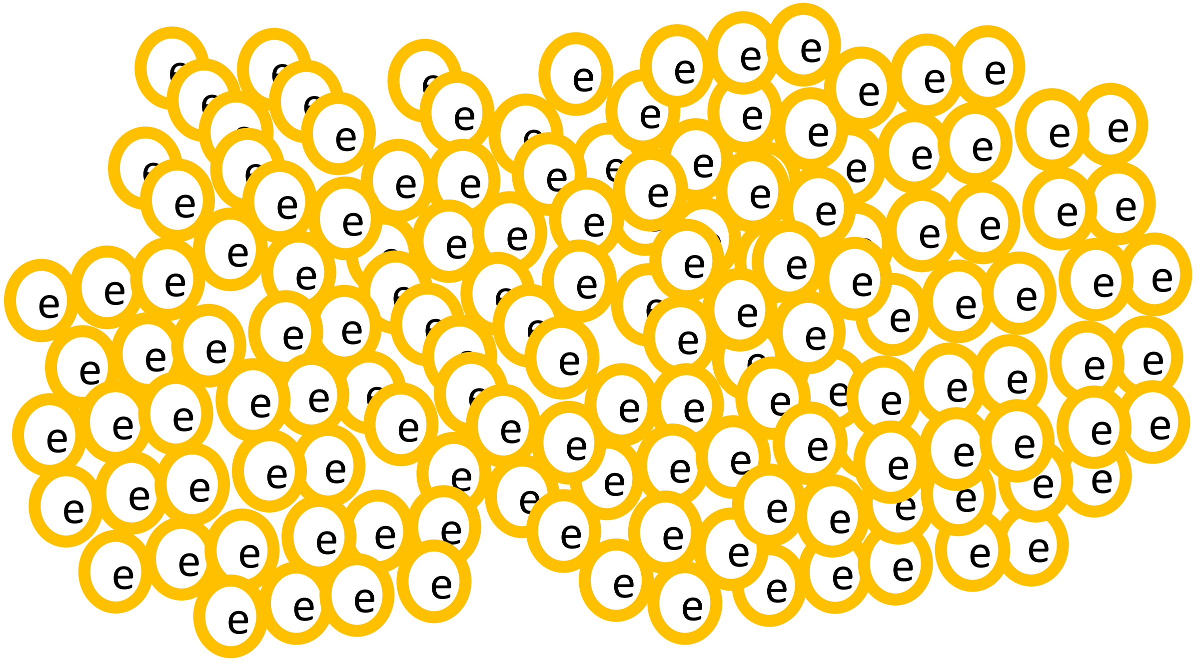
Electrons are tiny and there are so many flowing it makes no sense to count them so instead we use the word coulombs.



Electrical current is:

coulombs per unit of time”

1 coulomb = 6 241 500 000 000 000 000 electrons



Charge flow is measured in coulombs (C)

Higher Tier

- ❑ Recall
- ❑ Use
- ❑ Rearrange

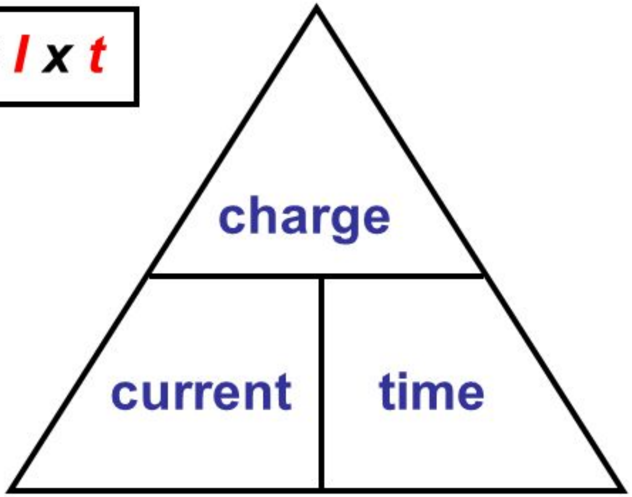
$$Charge\ flow\ (Q) = current\ (I)\ x\ time\ taken\ (t)$$

$$(coulombs, C) \quad = (amperes, A) \quad (seconds, s)$$

Foundation Tier

- ❑ Aided recall
- ❑ Use
- ❑ Rearrange

$Q = I \times t$

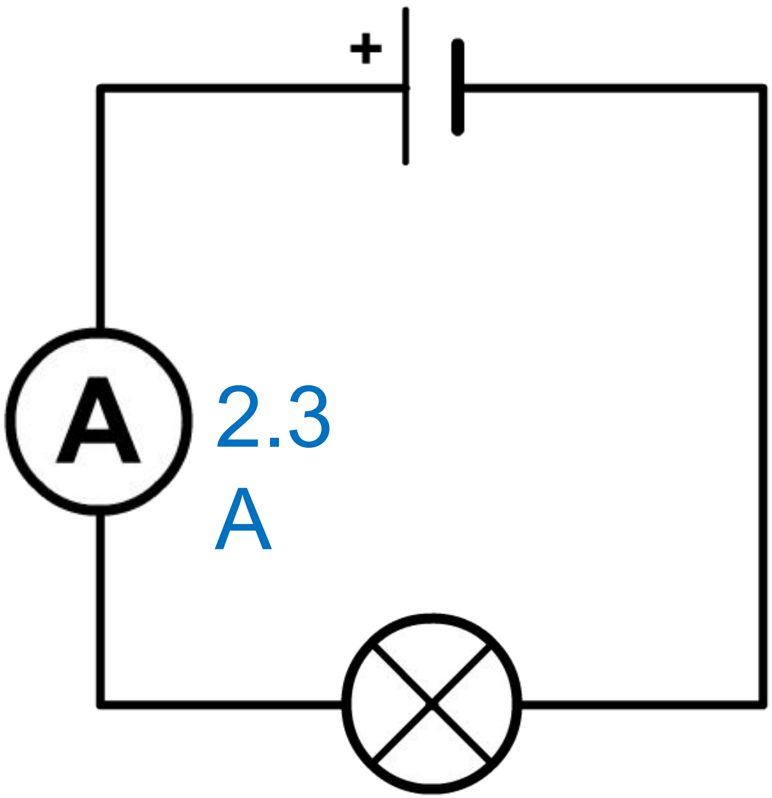


Worked Example:

An electric circuit had a current of 2.3 Amps. The time taken was 0.5 seconds. What was the charge flow

Foundation Tier

$$Charge\ flow\ (Q) = current\ (I)\ x\ time\ taken\ (t)$$



$$Charge\ flow\ = 2.3\ x\ 0.5$$

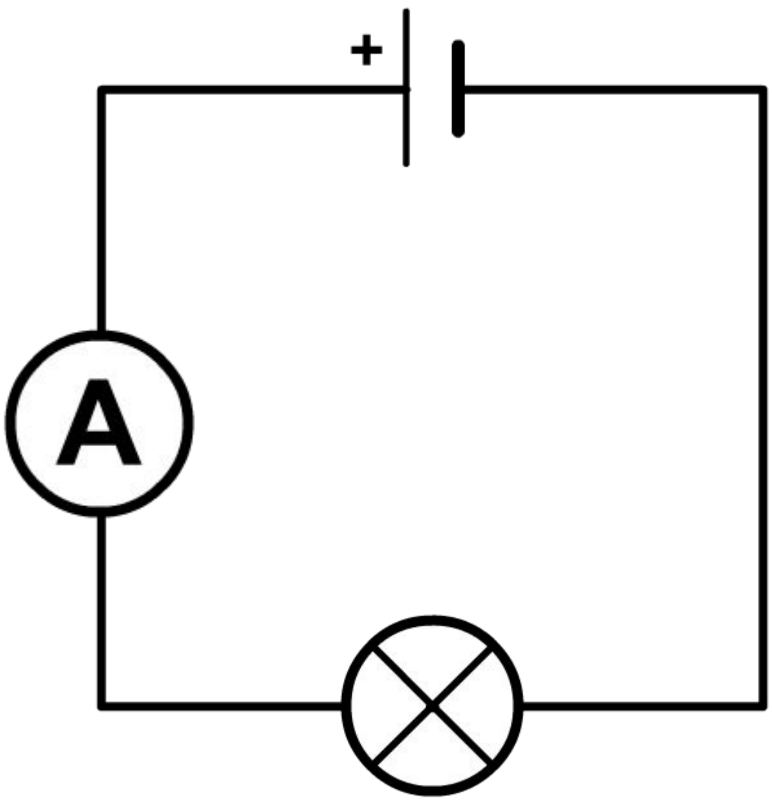
$$Charge\ flow\ = 2.3\ x\ 0.5$$

Worked Example:

An electric circuit had a charge flow of 25 coulombs. The time taken was 2.5 seconds. Calculate the current.

Higher Tier

$$Charge\ flow\ (Q) = current\ (I)\ x\ time\ taken\ (t)$$



$$current = \frac{charge\ flow}{time\ taken}$$

$$Charge\ flow = 2.3\ x\ 0.5$$

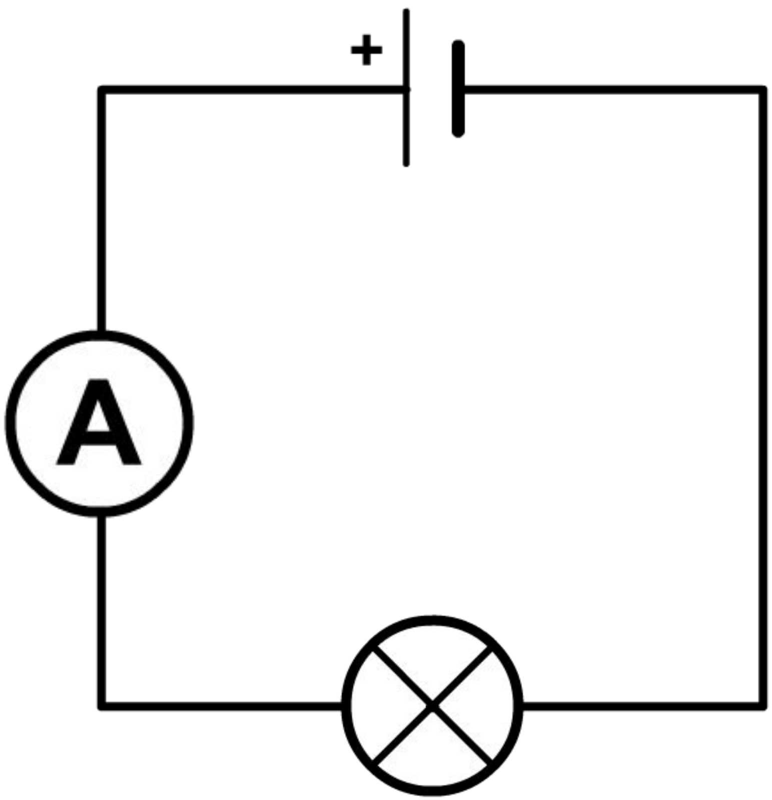
$$Charge\ flow\ = 2.3\ x\ 0.5$$

Worked Example:

An electric circuit had a charge flow of 2430 coulombs. The current was 33.23 amperes. Calculate the time to 3 S.F.

Higher Tier

$$Charge\ flow\ (Q) = current\ (I)\ x\ time\ taken\ (t)$$



$$time\ taken = \frac{charge\ flow}{current}$$

Charge flow = 2.3 x 0.5

Charge flow = 2.3 x 0.5

Activity: Complete these questions

TASK 3: Complete these practice foundation questions.

1.. What is the equation that links time taken, charge flow and current.

....[1]

2.. The current through a circuit was 24 A. The time taken was 3.6 s. Calculate the charge flow.

....[2]

3.. The current through a circuit was 12 A. The time taken was 0.5 s. Calculate the charge flow.

....[2]

4.. The current through a circuit was 15.25 A. The charge flow was 20 C. Calculate the time taken to 2 significant figures.

....[2]

Activity: Complete these questions

TASK 3: Complete these practice higher questions.

1.. What is the equation that links time taken, charge flow and current.

....[1]

2.. The current through a circuit was 56.2 A. The time taken was 2.3 s. Calculate the charge flow to 3 S.F.

....[2]

3.. The charge flow was 35 C. The time taken was 12.3 s. Calculate the current to 2 significant figures.

....[2]

4.. The charge flow was 35 C. The current was 7.23 A. Calculate the time taken to 2 D.P.

....[2]

5.. The charge flow was 120.2 C. The time taken was 8.8 s. Calculate the current to 1 decimal places.

....[2]

Review

1.. What is the equation that links time taken, charge flow and current.

Charge flow = current x time taken [1]

2.. The current through a circuit was 24 A. The time taken was 3.6 s. Calculate the charge flow.

24 A x 3.6 s = 84.6 C [2]

3.. The current through a circuit was 12 A. The time taken was 0.5 s. Calculate the charge flow.

12 A x 0.5 s = 6 C [2]

4.. The current through a circuit was 15.25 A. The charge flow was 20 C. Calculate the time taken to 2 significant figures.

Time taken = current / charge flow [1]   15.25 / 20 = 0.76 s [1]

Review

TASK 3: Complete these practice higher questions.

1.. What is the equation that links time taken, charge flow and current.

Charge flow = current x time taken [1]

2.. The current through a circuit was 56.2 A. The time taken was 2.3 s. Calculate the charge flow to 3 S.F.

56.2 x 2.3 = 129.26 [1]   129 C [1]

3.. The charge flow was 35 C. The time taken was 12.3 s. Calculate the current to 2 significant figures.

35 x 12.3 = 430.5 [1]   430 C [1]

4.. The charge flow was 35 C. The current was 7.23 A. Calculate the time taken to 2 D.P.

7.23 / 35 = 0.206 [1]   0.21 s [1]

5.. The charge flow was 120 C. The time taken was 8.8 s. Calculate the current to 1 decimal places.

120.2 x 8.8 = 1057.76 A [1]   1057.8 A [1]

LOb: Understand the terms current and charge.

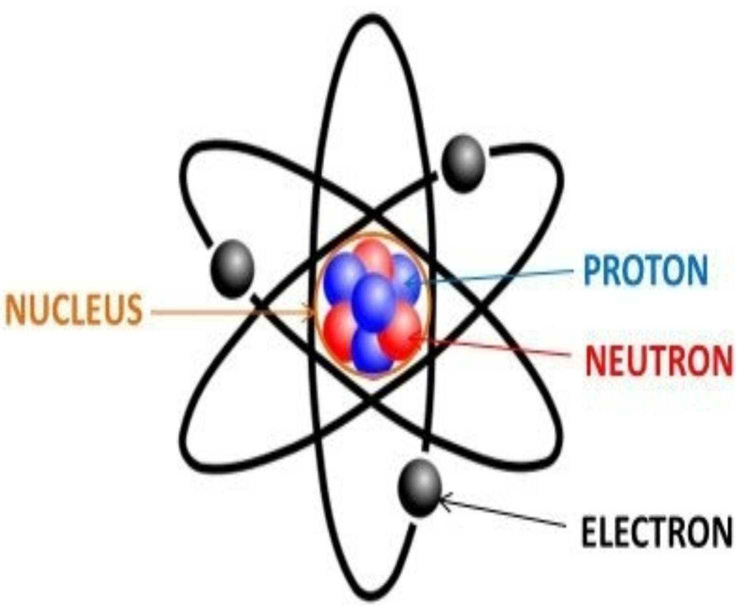
Keywords: Current, charge, coulombs, component, electrons

Learning outcomes:

✔Recall how electric circuits are shown as diagrams

✔Describe the current in an electric circuit

✔Calculate the size of an electric current



Did you know… electricity is all to do with how electrons move through wires.

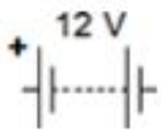
keyfacts

- **Each component that is added to a circuit has a circuit symbol**
- **Current** is a **flow of charge**. A flow of charge is a flow of **electrons** which are **negatively charged.**
- The equation for charge flow is
  Charge flow (coulombs) = current (amps) x time taken (seconds)
- 1 amp is equivalent to **1 coulomb per second.**

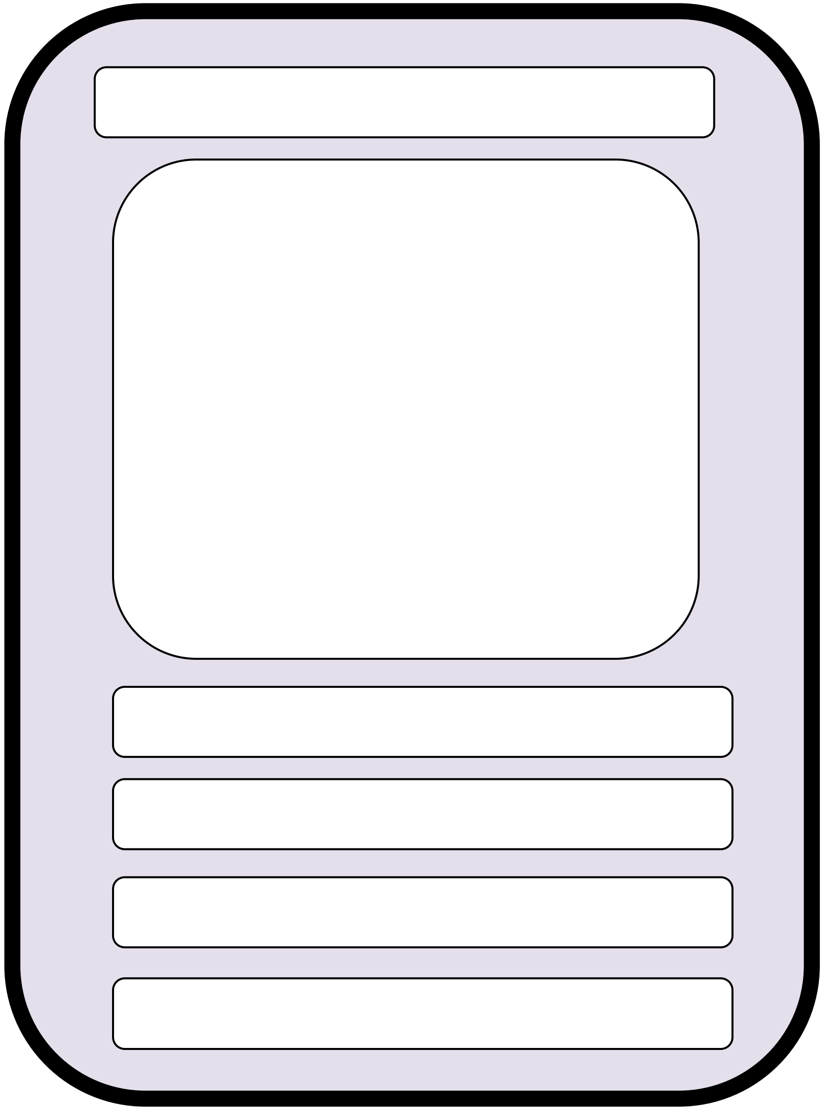
Plenary: Exam question

A student wanted to investigate the current flowing through a filament lamp with different levels of resistance added to the circuit.

Draw the circuit that they would construct to do this.

12 V battery	variable resistor	filament lamp	voltmeter	ammeter
				

Plenary



One key word that is stuck in your mind from today

A picture to describe todays learning

A sentence to describe
your learning today

	A surplus or deficit of negatively charged electrons
	A flow of charge. Number of electrons flowing through wires
	Something added to a circuit that affects the charge (bulb, motor, buzzer)
	The difference of electrical potential between two points.
	A flow of electric charge
	Equipment used to measure the amount of current
	Equipment used to measure the amount of voltage
	A measure of how current is prevented from flowing
	A circuit containing only one loop
	A circuit containing more than one loop

Charge	A surplus or deficit of negatively charged electrons
Current	A flow of charge. Number of electrons flowing through wires
Component	Something added to a circuit that affects the charge (bulb, motor, buzzer)
Potential difference	The difference of electrical potential between two points.
Current (amps)	A flow of electric charge
Ammeter (A)	Equipment used to measure the amount of current
Voltmeter (V)	Equipment used to measure the amount of voltage
Resistance	A measure of how current is prevented from flowing
Series circuit	A circuit containing only one loop
Parallel circuit	A circuit containing more than one loop